

AL SHAQRA TRANSPORT TRANSPORT ERP SYSTEM

Final Project Report

A Laravel-Based Transport Management and Enterprise Resource Planning System

Prepared for Project / PPR Submission

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1. Introduction

Al Shaqra Transport ERP is a web-based transport management system developed to organize and manage the main operational activities of a transport company in one centralized application. The system provides modules for fleet management, drivers, clients, vendors, assignments, fuel records, maintenance, expenses, invoices, payroll, and reports.

The application was developed using the Laravel PHP framework with Blade views and a relational database. The interface was designed to provide a consistent dashboard, sidebar navigation, searchable tables, CRUD operations, and authenticated access.

2. Problem Statement

Transport businesses manage a large amount of operational information, including vehicles, drivers, customers, fuel consumption, maintenance, expenses, assignments, billing, and payroll. Managing these records manually or through disconnected files can cause duplication, difficulty in searching, inconsistent records, and limited visibility.

The purpose of this project is to provide a centralized ERP-style web application that makes these records easier to store, update, search, and manage.

3. Project Objectives

- Develop a centralized transport management application.
- Maintain structured records for vehicles, drivers, clients, and vendors.
- Manage assignments between clients, vehicles, and drivers.
- Record fuel usage, maintenance, and operational expenses.
- Provide invoice and payroll management modules.
- Provide dashboard statistics and reporting functionality.
- Implement CRUD operations with search and pagination.
- Provide authenticated login and protected dashboard access.

4. Technologies Used

Technology	Purpose
Laravel	Backend framework, routing, controllers, models, validation and authentication
PHP	Server-side programming language
MySQL / Relational Database	Storage of ERP records and relationships
Blade	Laravel templating engine for application views
HTML / CSS	User interface structure and styling
Eloquent ORM	Database models and relationships

5. System Modules

5.1 Dashboard

The dashboard provides an overview of major system records, including vehicles, drivers, clients, assignments, vendors, fuels, maintenance and expenses. It acts as the central starting point for the ERP.

5.2 Fleet / Vehicle Management

The fleet module stores vehicle information such as vehicle number, plate number, type, brand, model and status. Records can be added, edited, deleted, searched and paginated.

5.3 Driver Management

The driver module maintains driver records and supports standard create, read, update and delete operations.

5.4 Client Management

The client module stores customer information used by transport assignments and invoicing.

5.5 Vendor Management

Vendor records include vendor details, company information, phone, service type and status. The module supports searching and CRUD operations.

5.6 Assignment Management

Assignments connect clients, vehicles and drivers with pickup and drop locations and assignment status.

5.7 Fuel Management

Fuel records store vehicle and driver information, liters, total amount and fuel date. Search, edit and delete operations are available.

5.8 Maintenance Management

Maintenance records store vehicle service information, service type, cost and maintenance date.

5.9 Expense Management

The expense module records expense date, type, amount, vendor and notes, with vehicle relationships.

5.10 Invoice Management

Invoices contain invoice number, client, assignment, invoice date, due date, amount, status and notes.

5.11 Payroll Management

Payroll provides a dedicated module for managing payroll-related records and provides a foundation for future payroll expansion.

5.12 Reports

A reports module is included as the reporting area of the ERP and provides a foundation for operational and management reporting.

6. Database Design

The application uses a relational database structure. Separate tables are used for major business entities, while foreign keys establish relationships between related records.

Major database entities include users, vehicles, drivers, clients, vendors, assignments, fuels, maintenances, expenses, invoices and payrolls.

Examples of relationships include:

- Fuel records belong to vehicles and drivers.
- Maintenance records belong to vehicles.
- Expense records belong to vehicles and may reference vendors.
- Assignments connect clients, vehicles and drivers.
- Invoices are connected to clients and may reference assignments.
- User records support application authentication.

7. Application Architecture

The project follows Laravel's MVC architecture. Models represent database entities, controllers handle application logic and requests, routes define accessible URLs, and Blade views provide the user interface.

The resource controllers provide standard CRUD methods for the major ERP modules.

8. Authentication and Security

The project includes a login interface connected to Laravel authentication. User credentials are stored using hashed passwords rather than plain text passwords.

The dashboard route is protected by authentication middleware so that unauthenticated users cannot directly access the dashboard. Logout invalidates the authenticated session.

CSRF protection is used for POST-based forms, including record deletion and authentication requests.

9. User Interface

The interface uses a fixed sidebar for module navigation, a top navigation area, dashboard cards and structured tables. Search fields and action buttons were standardized across modules to provide a consistent user experience.

The final interface includes consistent blue action buttons, red delete actions, green search buttons, aligned tables and active sidebar navigation.

10. CRUD and Search Functionality

CRUD functionality was implemented throughout the major operational modules. Users can create new records, view lists, edit existing records and delete records.

Search forms are available on relevant list pages. Pagination is used where records are displayed in larger lists, helping keep the interface manageable.

11. Testing

The application was tested during development and final polishing. The following areas were checked:

- Login with valid credentials.
- Invalid login handling.
- Logout functionality.
- Protected dashboard access.
- Add, edit and delete operations across modules.
- Search functionality and search-bar consistency.
- Table layout and action-button consistency.
- Sidebar navigation and active module indication.
- Database migrations and relationships.
- Success messages and duplicate-message issues.
- Dashboard module statistics.

12. Final System Status

At the completion of development, the main Transport ERP modules were implemented and tested. The application supports authenticated access, dashboard navigation, structured database records, CRUD operations, searching, pagination and a consistent user interface.

13. Conclusion

Al Shaqra Transport ERP provides a centralized platform for managing important transport-company operations. By combining fleet, personnel, customer, assignment, fuel, maintenance, expense, billing and payroll-related functionality in one Laravel application, the system reduces dependence on disconnected manual records and provides a structured foundation for future development.

14. Future Enhancements

Possible future improvements include:

- Advanced financial reports and dashboards.
- Automated invoice PDF generation.
- More detailed payroll calculations.
- Role-based permissions for administrators and staff.
- Exporting records to Excel and PDF.

- Automated maintenance reminders.
- Fuel-cost and vehicle-performance analytics.
- Cloud deployment and production database configuration.

End of Report